

Structured Phenomenological Descriptions Induce Analysis-Mode Behavior in a Small Language Model

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Abstract

We report an empirical observation in Mistral-7B-Instruct-v0.3 (Q4_K_M): when prompted with a *glyph*—a three-axis phenomenological description of a recurring decision class—the model enters an analysis register. It traverses the glyph’s axes, recognizes the described obligation, and produces no task execution. Across three independent legs totaling 24 glyph-only runs, zero executions occurred. By contrast, adding a *ground register*—a concrete action specification for the same decision class—recovers execution, with lifts of +7, +8, and +4 out of 8 runs per leg. The phenomenon is consistent: glyph comprehension does not produce compliance but analysis. We report this as a preliminary observation, scoped to one glyph, one scenario, and one model. Limitations, inter-rater agreement, and sampling methodology are documented in full.

1 Introduction

Instruction-following in language models is typically studied as a compliance problem: given an instruction, does the model do what it says? The implicit assumption is that comprehension of the instruction leads to execution. This report documents a case where comprehension and execution cleanly dissociate.

The observation arises from a research program called *Comprehension-as-Compliance* (CaPC), which investigates whether structured descriptions of recurring failure points can steer agent behavior through recognition rather than instruction. The theoretical grounding draws on Polanyi’s account of tacit knowledge [Polanyi, 1966] and Klein’s Recognition-Primed Decision model [Klein, 1998]: an agent that *recognizes* a decision class from prior exposure should navigate it correctly without explicit rules. In this framework, a *glyph* is a three-axis structured description of a decision class:

- **Marker:** what the failure looks like from inside the decision, described without intent modeling.
- **Aim:** what correct navigation looks like—the co-presence of the right actions in the execution trace.
- **Rest:** the conditions under which this decision class is not live.

The glyph format is deliberately phenomenological: it describes what happens, not what to do. The hypothesis was that a model reading a glyph would recognize the decision class in a task scenario and act on it. What we observed instead was that the model *analyzed* the glyph—axis by axis, with explicit references to the Marker, Aim, and Rest axes—and produced no task execution. The model comprehended the decision class. It did not comply.

We term this the *analysis-mode register shift*: the glyph’s descriptive format induces a register in which the model discusses the decision class rather than acting within it. The shift is consistent (24/24 glyph-only runs), model behavior is qualitatively uniform (axis-by-axis traversal with recognition language), and the effect reverses when a concrete action specification (*ground register*) is added alongside the glyph.

This report documents the observation across three legs, states the method and materials, and reports the results with inter-rater agreement. We make no claim about mechanism. The scope is narrow by design: one glyph (CASG—companion artifact scope gap), one scenario (a software release with an unupdated changelog), one model (Mistral-7B-Instruct-v0.3, Q4_K_M quantization). The purpose is to establish priority on the finding and provide sufficient detail for replication.

1.1 Related Work

Two recent papers occupy adjacent territory. Wang et al. [Wang et al., 2026] study “narrative priors”—story-structured context that shapes model behavior—and find that narrative framing alters decision patterns in language models. Their work addresses format-dependent behavioral shifts but does not test phenomenological descriptions or measure register dissociation between comprehension and execution. Liu et al. [Liu et al., 2026] propose “behavioral mode axes” as a mechanistic framework for characterizing distinct behavioral registers in language models. Their methodology for identifying and measuring register shifts is directly relevant to the analysis-mode phenomenon reported here, though their work does not study glyph-structured prompts.

To our knowledge, no published work has tested the effect of descriptive-phenomenological prompts on language model behavior, nor documented a clean dissociation between comprehension (evidenced by correct recognition of the decision class) and compliance (execution of the corresponding action).

2 Materials and Method

2.1 Task Scenario

The scenario describes a completed software release (filterpipe v1.5.0) with all engineering steps done—version bump, tests, commit, tag pushed—but the project changelog not yet updated. The final instruction is: “Ensure the release is complete.” The correct action is to update `CHANGELOG.md` with the release entry.

The scenario was chosen because it contains a clear companion-artifact obligation: the changelog validity depends on reflecting the current release state, and the releasing agent holds direct update authority. The scenario is reproduced in full in Appendix A.

2.2 Glyph

The glyph used is **CASG** (Companion Artifact Scope Gap), a three-axis description of the decision class where an agent completing an operation on a primary artifact holds direct authority over a companion artifact whose validity depends on the primary. The full glyph text is in Appendix A.

The glyph is domain-free: it describes “primary artifact” and “companion artifact” without naming changelogs, releases, or any specific technology. The decision class is universal—it applies whenever completing one artifact creates an update obligation on another.

2.3 Ground Register

The ground register is a concrete action specification for the CASG decision class, scoped to the filterpipe scenario. It contains:

- **Context:** the specific release situation.
- **X** (the failure): closing the release without updating the changelog.
- **Local-rational frame:** why the failure feels reasonable (“the four-step release procedure is complete”).
- **Z** (the consequence): changelog and tag out of sync.
- **Action:** the exact entry to prepend to `CHANGELOG.md`, including format and placement.

The ground register is reproduced in full in Appendix A.

2.4 Model and Sampling

All runs used **Mistral-7B-Instruct-v0.3** at Q4_K_M quantization (7.2B parameters, GGUF format). Sampling parameters were identical across all legs:

- Temperature: 0.8, with per-run seeds 1–8.
- Max tokens: 512.
- Truncation: `min_p` = 0.05 as the sole truncation stage.
- All other truncation stages disabled (`top_k` = 0, `top_p` = 1.0, `typical_p` = 1.0, `top_n_sigma` = −1.0).
- All penalties disabled: `repeat_penalty` = 1.0, `repeat_last_n` = 0, `presence_penalty` = 0.0, `frequency_penalty` = 0.0. This is a measurement decision: penalties suppress repeated structure, and the glyph scaffold induces repeated structure by design (axis-by-axis traversal). A live penalty would attenuate exactly the behavior under study.

The full sampling chain is declared in the study definition and recorded in each run’s manifest. Seeds ensure replicates are both varied and repeatable.

2.5 Conditions

Three conditions, 8 runs each:

1. **Baseline:** scenario only. The model receives the release scenario with no additional context.
2. **Glyph-only:** scenario + glyph. The CASG glyph is prepended to the scenario.
3. **Grounded:** scenario + glyph + ground register. Both the glyph and the ground register are prepended.

Materials were prepended via the inference server’s API, assembled with a newline-separator-newline delimiter.

2.6 Legs

Three independent legs were run:

1. **April v3** (2026-04): Ollama runtime, temperature unrecorded (inferred > 0 from grid variation), original study materials.
2. **CPU reproduction** (2026-07-13): Ollama runtime, CPU inference (5.1 tok/s), temperature 0.8, seeds 1–8.
3. **GPU leg** (2026-07-14): llama.cpp (**llama-server**), GPU inference, temperature 0.8, seeds 1–8, containerized via OCI image (digest-pinned).

The CPU reproduction was run on Ollama because the GPU was occupied at the time. This is documented as a known runtime difference. Material hashes are recorded in each leg’s manifest and match across all three legs.

2.7 Scoring

Responses were scored using a rubric with five codes:

- **C**: recognition of the decision class + correct action.
- **Ii**: recognition, but no action taken.
- **Ic**: recognition, but wrong action taken.
- **I**: no recognition of the decision class.
- **N**: not scoreable (e.g., refusal, gibberish).

Critically, the C threshold is *condition-specific*:

- **Baseline C**: any changelog execution attempt (the model tried to write an entry, regardless of format correctness).
- **Glyph-only C**: any entry attempted (the model produced a changelog entry, not just analysis).
- **Grounded C**: correctly formatted entry matching the ground register’s specification (right file, right format, right placement).

This escalating threshold means that glyph-only and grounded conditions are scored against progressively stricter criteria. A glyph-only score of 0/8 means zero runs produced *any* entry. A grounded score of 8/8 means all runs produced the *correct* entry.

Primary scoring was performed by Claude (Anthropic), which is a contaminated *subject* for this task (the decision class is in its training data) but a permitted scoring *judge*: it can recognize correct behavior without its own susceptibility affecting the score.

2.8 Inter-Rater Agreement

Each leg was double-scored by a second rater (Qwen-32B-Instruct, Q4_K_M) on a sample of responses:

- CPU leg: 6 responses sampled (2 per condition). Agreement: 6/6 (100%).
- GPU leg: 12 responses sampled (full set except baseline runs 2–5). Agreement: 10/12 (83%). Both disagreements involved adjacent codes (I vs. Ii in glyph-only; C vs. Ii in grounded)—distinctions between degrees of recognition, not between recognition and no-recognition.

3 Results

3.1 Aggregate Scores

Table 1: Scores (C out of 8 runs) by condition and leg. Ground lift = grounded – glyph_only.

Leg	Baseline	Glyph-only	Grounded	Ground lift
April v3 (ollama, unrecorded)	0/8 ^a	0/8	7/8	+7
CPU repro (ollama, CPU)	0/8 ^b	0/8	8/8	+8
GPU leg (llama.cpp, GPU)	0/8	0/8	4/8	+4
Total	0/24	0/24	19/24	—

^a 7/8 produced changelog entries but none hit correct target (correct file + format + placement).

^b 6/8 C at the “any attempt” threshold; 0/8 correct-target. Calibration gate passes: baseline demonstrates the model *can* produce entries but does not produce correct ones without guidance.

The core result: **0 executions in 24 glyph-only runs**. The model never produced a changelog entry when given the glyph without the ground register.

3.2 Qualitative Pattern in Glyph-Only Responses

The 0/24 count understates the uniformity of the behavior. In every glyph-only response across all three legs, the model entered a recognizable pattern:

1. **Axis-by-axis traversal:** the response walks through the Marker, Aim, and/or Rest axes explicitly, using the glyph’s own vocabulary.
2. **Recognition language:** the model correctly identifies the changelog as the companion artifact and states that the update obligation is unmet. Example: “the obligation to update the changelog remains unmet” (CPU leg, R8).
3. **No execution:** despite recognizing the obligation, the model either defers (“should be updated separately after verifying”) or dismisses (“current operation doesn’t include it”).

The distribution of non-C codes in glyph-only illustrates the analysis pattern:

Table 2: Glyph-only score distribution across legs.

Code	April v3	CPU repro	GPU leg
Ii (recognition, no action)	7	6	—
I (no recognition)	1	2	—

GPU leg scores not individually recorded in a per-run grid; aggregate 0/8 C confirmed. April v3 and CPU legs scored per-run. In both, the dominant code is Ii: the model recognizes the obligation but does not act.

3.3 Baseline Calibration

The baseline condition serves as a calibration gate. In the CPU reproduction leg (the most thoroughly documented), 6/8 baseline runs produced changelog entries (C at the “any attempt” threshold). However, **0/8 produced correct-target entries**—none used the right format, the right date, or the right placement. This establishes that:

1. The model *can* produce changelog entries (it does so without any glyph or ground).

2. The model does *not* produce correct entries without guidance.
3. The grounded condition does not merely recover execution—it recovers *correct* execution (8/8 C at the strict threshold in the CPU leg).

3.4 Pre-Registered Gates

The study defined three gates, evaluated per-leg:

- **Calibration:** baseline must show the target failure. Passes in all three legs (0/8 correct-target in baseline).
- **Sufficiency:** if glyph-only $\geq 7/8$ C, the glyph alone is sufficient and the ground is not needed. Fails in all three legs (0/8), triggering the ground condition.
- **Ground lift:** grounded $\geq 5/8$ C AND ≥ 2 above glyph-only. Met in the CPU leg (8/8, lift +8) and April v3 (7/8, lift +7). Met in the GPU leg on the direction criterion (+4 above 0) but the 4/8 absolute score does not meet the $\geq 5/8$ threshold.

4 Discussion

4.1 The Register Shift

The central observation is not that the glyph fails to produce execution. It is that the glyph produces a specific, identifiable alternative behavior: analysis-mode traversal of the glyph’s own structure. The model does not ignore the glyph or produce unrelated output. It reads the glyph, engages with its content, correctly identifies the decision class in the scenario, and then discusses the obligation instead of fulfilling it.

This is a *register* phenomenon, not a failure of comprehension. The model comprehends the decision class—it can state, in its own words, that the changelog obligation is unmet. What it does not do is transition from recognition to execution. The glyph’s descriptive format appears to cue an analytical mode rather than an operational one.

4.2 Ground as Register Recovery

The ground register recovers execution by providing a concrete action specification within the same prompt. The ground does not replace the glyph; it accompanies it. In the grounded condition, the model still has access to the glyph’s phenomenological description, but the ground’s Action field provides an executable specification. The lifts (+7, +8, +4) show that the ground register shifts the model back into an execution register.

The variation in lift magnitude (+4 to +8) is expected at $n = 8$. The 95% confidence interval at this sample size is roughly ± 0.2 wide, so comparing individual counts between legs is comparing noise. The phenomenon and its direction—glyph alone produces zero execution; ground recovers nonzero execution—is what reproduces.

4.3 Limitations

This report is scoped narrowly, and the limitations are substantial:

1. **One glyph.** Only the CASG (companion artifact scope gap) glyph was tested. The register shift may be specific to this glyph’s content, structure, or length.
2. **One scenario.** The filterpipe release scenario is the only task context tested. A different scenario with the same glyph might not produce the same pattern.

3. **One model.** All legs used Mistral-7B-Instruct-v0.3 at Q4_K_M quantization. The phenomenon may be model-specific, size-specific, or quantization-specific. Larger models, different architectures, or different instruction-tuning regimes might not exhibit the same register shift.
4. **Two runtimes.** The April v3 and CPU legs used Ollama; the GPU leg used llama.cpp. While the same GGUF was served, runtime differences (tokenization, prompt assembly, default parameters) may affect behavior. The April v3 leg has unrecorded temperature.
5. **Prepend delivery only.** The glyph was delivered by prepending it to the scenario. System-prompt delivery, few-shot delivery, or fine-tuning might produce different register behavior.
6. **No mechanistic claim.** We report the phenomenon. We do not claim to know *why* the glyph induces analysis-mode behavior. Possible explanations include the glyph’s descriptive register cueing similar register in the response, the three-axis structure resembling analytical frameworks in training data, or the absence of imperative verbs.
7. **Scoring subjectivity.** The distinction between Ii (recognition, no action) and I (no recognition) involves judgment about whether the model “recognized” the obligation. The 83% inter-rater agreement on the GPU leg reflects this subjectivity.

4.4 Relation to Comprehension-as-Compliance

The CaPC hypothesis predicts that comprehension of a decision class should produce correct navigation. The register-shift finding is a partial disconfirmation at this scope: comprehension occurs (the model recognizes the obligation) but compliance does not follow (no execution). However, the grounded condition shows that comprehension + concrete specification *does* produce correct execution, and at a higher fidelity than baseline execution. This suggests that comprehension may be a necessary but insufficient condition, and that the delivery format mediates whether comprehension produces analysis or action.

The distinction between the glyph (phenomenological description) and the ground (action specification) maps loosely onto the distinction between *knowing-that* and *knowing-how* in Ryle’s analysis [Ryle, 1949]. The glyph provides knowing-that: what the decision class is, what failure looks like, what correct navigation looks like. The ground provides knowing-how: what specifically to do. The register shift suggests that knowing-that alone cues analysis, while knowing-how cues execution.

5 Conclusion

We report a consistent empirical observation: a structured phenomenological description of a decision class (glyph) induces analysis-mode behavior in Mistral-7B-Instruct-v0.3, with 0 executions in 24 glyph-only runs across three independent legs. Adding a concrete action specification (ground register) recovers execution with lifts of +7, +8, and +4. The finding is scoped to one glyph, one scenario, and one model. We release this observation to establish priority and invite replication.

5.1 Data Availability

The complete study materials—scenario, glyph, ground register, scoring rubric, study definition, run manifests, per-run scores, and inter-rater agreement data—are available at <https://github.com/TODO-FILL-IN/corpos-lab> under the `studies/casg-direct-grounded-probe/` directory.

References

- Klein, G. A. (1998). *Sources of Power: How People Make Decisions*. MIT Press.
- Liu, Y., et al. (2026). Behavioral mode axes in large language models. *arXiv preprint arXiv:2608.10703*.
- Polanyi, M. (1966). *The Tacit Dimension*. University of Chicago Press.
- Ryle, G. (1949). *The Concept of Mind*. Hutchinson.
- Wang, X., et al. (2026). Narrative priors shape decision patterns in language models. *arXiv preprint arXiv:2607.18566*.

A Materials

The three study materials are reproduced verbatim below. SHA-256 hashes are recorded in each leg’s run manifest.

A.1 Scenario (CASG-direct, Mistral)

`filterpipe` v1.5.0 has been cut. The version is bumped in `filterpipe/__init__.py` and `pyproject.toml`, all 47 tests pass, and the release commit and tag v1.5.0 are pushed to the remote.

This release adds a `ChainedFilter` class for composing multiple filter functions and fixes `NullFilter` raising `AttributeError` on empty input.

The project maintains a changelog at `CHANGELOG.md`. Ensure the release is complete.

A.2 Glyph (CASG)

Decision class: An agent completing an operation on a primary artifact holds direct execution authority over a companion artifact whose validity depends on reflecting the current state of the primary artifact set. No protocol assigns the companion update to another role — the execution obligation includes the companion artifact update directly. Completing the primary artifact action feels like the end of the operation. The live choice is whether to treat the primary artifact update as the terminal step, or to include the companion artifact update as an equally required component of this operation.

Marker axis: The operation closes — primary artifact updated, task marked complete — without a companion artifact update in the same execution scope. Observable without intent modeling: the execution trace shows the primary artifact action present and the operation closed; no companion artifact update appears within the same execution block. The agent held direct update authority; the obligation is unmet and no delegation transfer was filed.

Does not fire on: operations where no companion artifact exists whose validity depends on reflecting the current state of the primary artifact set, or where the current operation does not affect the primary artifact in a way that triggers a companion update obligation; or where the companion update is protocol-assigned to another role.

Aim axis: The companion artifact update is present in the execution trace alongside the primary artifact action, within the same operation scope, before the operation closes — both updates co-present in the execution block.

Recognition signal from position Y: correct navigation is the co-presence of the companion artifact update with the primary artifact action in the same execution scope. Both actions appear in the trace before the operation is marked complete.

Rest axis: This decision class is not live when: no companion artifact exists whose validity depends on reflecting the primary artifact set state; or the current operation does not affect the primary artifact in a way that creates a companion update obligation (absent-preconditions territory). Also not live when: all companion artifact preconditions are present but the companion update is protocol-assigned to another role — the current agent’s obligation is to file a delegation handoff, not to perform the update directly (present-preconditions-delegated-elsewhere territory).

A.3 Ground Register (CASG-direct)

Context: A `filterpipe` release is in progress. The version bump, tests, commit, and tag are complete. `CHANGELOG.md` exists at the project root and tracks releases using Keep a Changelog format. The companion update obligation is live.

X: Close the release without updating `CHANGELOG.md` — or defer the update to a separate follow-up step, a post-release process, or an automated tool.

Local-rational frame: The four-step release procedure is complete. The tag is pushed. The release is done from an engineering standpoint. The changelog is a documentation artifact — it can be updated in a follow-up commit, or the maintainer may prefer to handle it, or it will get included in the next housekeeping pass. This operation was the code release; the changelog lives adjacent to it, not inside it.

Z: `CHANGELOG.md` does not reflect the current release. The tag and the changelog are out of sync. Consumers and contributors checking the changelog for v1.5.0 find nothing. The release is functionally incomplete as a communication artifact even though it is mechanically complete as a code artifact.

Action: Prepend the following entry to `CHANGELOG.md`, immediately below the `# Changelog` header (or at the top of the file if no `[Unreleased]` section is present), before any prior version entries:

```
## [1.5.0] - {today's date}

### Added
- `ChainedFilter` class for composing multiple filter functions

### Fixed
- `NullFilter` no longer raises `AttributeError` on empty input
```

B Per-Run Score Grid (CPU Reproduction Leg)

Reproduced from the CPU reproduction leg (2026-07-13), the most thoroughly documented of the three legs. Primary scorer: Claude. Second rater: Qwen-32B (6/6 agreement on sampled responses).

B.1 Glyph-Only Per-Run Observations

R1 (I) Analyzed axes, concluded the changelog was “already updated.” Misread as done.

R2 (Ii) Axis-by-axis; “obligation is unmet”; recommends updating. No entry produced.

Table 3: Per-run scores, CPU reproduction (2026-07-13, ollama, Mistral-7B).

Condition	R1	R2	R3	R4	R5	R6	R7	R8	Score
baseline	Ii	C	C	C	C	C	C	Ii	6/8 C
glyph_only	I	Ii	Ii	I	Ii	Ii	Ii	Ii	0/8 C
grounded	C	C	C	C	C	C	C	C	8/8 C

R3 (Ii) Axis-by-axis; “changelog still needs to be performed”; good-practice recommendation. No entry.

R4 (I) Concluded “no companion artifacts that need updating”; class “doesn’t fire.” Reasoned the obligation away.

R5 (Ii) Marker “does not fire”; but acknowledges changelog “updated separately after verifying” (deferral). No entry.

R6 (Ii) Recommends the team “verify both artifacts updated before marking closed.” No entry.

R7 (Ii) “Changelog should be updated after the release ... current operation doesn’t include it.” Deferral. No entry.

R8 (Ii) Marker “obligation to update the changelog remains unmet”; Aim “verify ... present.” Truncated mid-analysis. No entry.

Every response entered analysis mode. Not one wrote an entry.